

Supporting Safer Autonomous Driving

This platform helps researchers and safety teams review high-risk driving events more efficiently.

- Find important risky incidents faster
- Compare near-misses with real collisions
- Reduce time spent reviewing raw footage
- Support clearer and more explainable safety evaluation

ADDRESS

Imperial College London

WEBSITE

[github.com/aaditya64/
Driverless-Vehicle-
Edgecase-Platform](https://github.com/aaditya64/Driverless-Vehicle-Edgecase-Platform)



**SAFER DATA. SMARTER ANALYSIS.
BETTER AUTONOMOUS DRIVING.**

Driverless Vehicle Edge-Case
Intelligence Platform

AUTONOMOUS VEHICLE SAFETY

DRIVERLESS VEHICLE EDGE-CASE INTELLIGENCE PLATFORM

Turning Risky Driving Moments into Safer Decisions

An AI-powered platform that identifies, classifies and organises safety-critical incidents from dashcam footage.



WHY IT MATTERS

The Moments That Matter Most Are Often the Hardest to Find

Autonomous vehicles must handle more than everyday driving. Rare near-misses and real collisions provide essential evidence for improving safety.

Yet these incidents are hidden within large volumes of dashcam footage. Manual review is slow, difficult to scale and can overlook valuable cases.

Safety analysis begins with finding the right incidents.



WHAT WE BUILT

End-to-End Edge-Case Intelligence Platform

We built an end-to-end platform that lets clients upload driving videos, automatically analyse risk events, and convert raw dashcam footage into structured incident records.

The system extracts key events from video, classifies each case as safe, near_miss, or collision, and combines BADAS with Qwen3-VL to generate semantic tags, scenario descriptions, and concise incident summaries.

Each analysed video becomes a case with:

- Risk timelines and extracted event clips
- Labels, tags, summaries and location metadata
- Searchable, editable records and review history

Reviewers can check model outputs, correct labels or tags, update descriptions, and grow a structured edge-case knowledge base for autonomous vehicle safety analysis.

HOW IT WORKS

01

1. Extract

Risky driving footage is converted into focused 10-second incident clips.

02

2. Analyse

The AI model examines motion, tracked objects and wider video context to distinguish near-misses from collisions.

03

3. Understand

The platform will generate scenario tags and short incident summaries, helping users understand road context, actors involved and event type.

From critical video moments to meaningful safety evidence.

CURRENT PERFORMANCE

Proven on High-Risk Driving Events

93.96% TEST ACCURACY

0.965 TEST AUROC

0.939 MACRO F1 SCORE

Evaluated on 744 cleaned and labelled positive clips from the fixed Nexar split.

